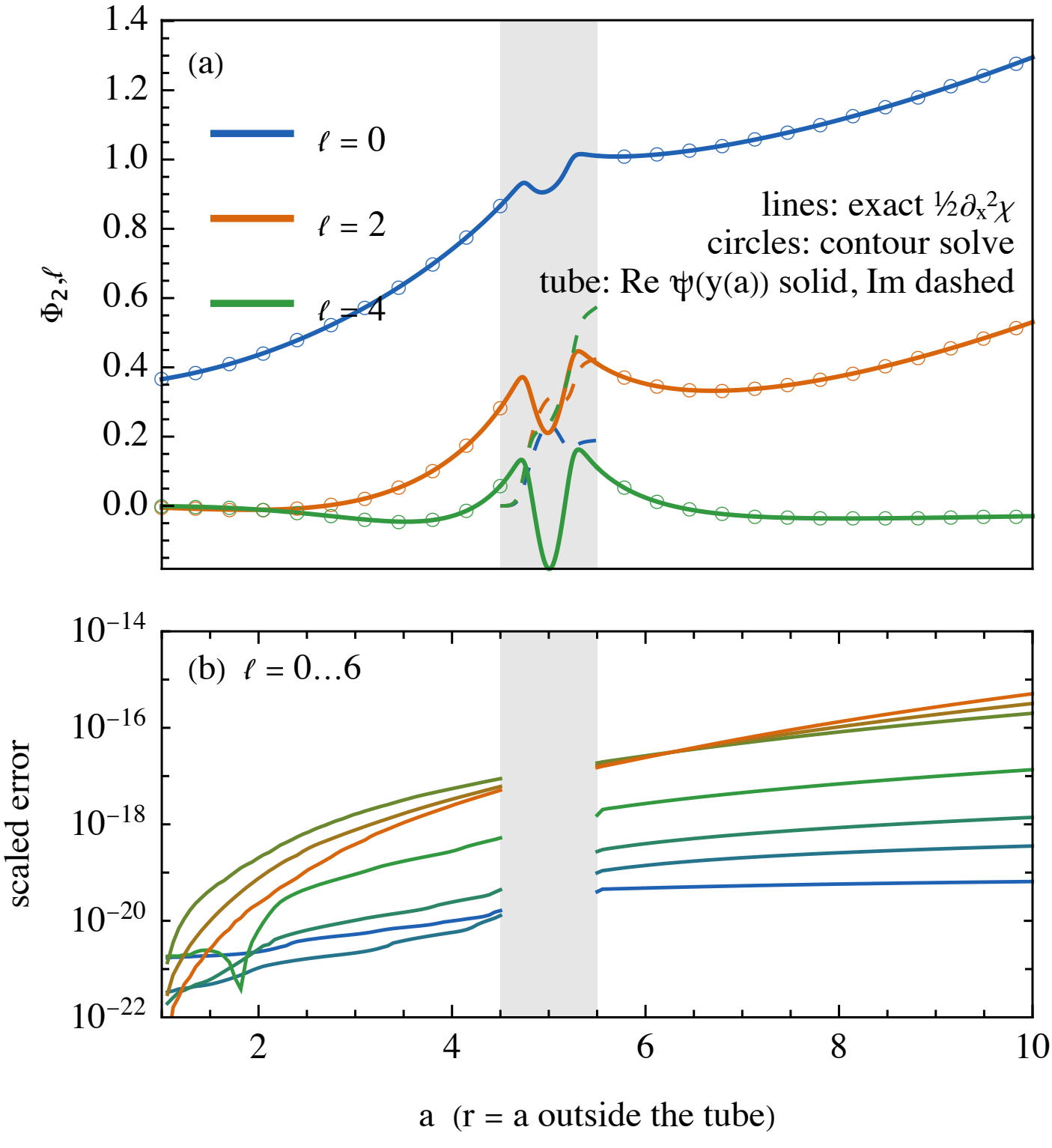
(a)
ℓ = 0
ℓ = 2
ℓ = 4
lines: exact ½∂x²χ
circles: contour solve
tube: Re ψ(y(a)) solid, Im dashed
Φ2,ℓ
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
(b) ℓ = 0…6
scaled error
10−14
10−16
10−18
10−20
10−22
2
4
6
8
10
a (r = a outside the tube)